

CIV 102 Team 209 Bridge Design Analyses and Construction Procedures

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CIV102 Team 209 Bridge Project Design Report

Member: Ethan L., Amy L., Xin Wang

Nov. 25th 2024

Fast and Easy: Super strength formula sets in seconds with maximum bond in 15-30 seconds with no clamping required

Versatile: Works well on a variety of porous and non-porous surfaces including leather, china, wood, rubber, metal, paper, ceramic, hard plastics and more



TECHNICAL DATA		
<u>Clean Up:</u>	Clean up uncured residue with acetone. Scrape away cured adhesive using a sharp-edged tool. Follow solvent manufacturer's precautions for using solvents.	
<u>Shelf Life:</u>	From date of manufacture (unopened): 18 months	<u>Lot Code Explanation:</u> XYDDD (printed on bottom of bottle) X = Disregard Y = Last digit of year of manufacture DDD = Day of manufacture based on 365 days in a year Example: A3061 = 61 st day of 2023 = March 2, 2023

Typical Cured Performance Properties		
<u>Color:</u>	Colorless to straw	<u>Service Temperature:</u> Up to 82°C (180°F)
<u>Cured form:</u>	Non-flammable, brittle solid	<u>Moisture Resistance:</u> Yes
<u>Tensile <u>Shear</u> Strength:</u>	Varies from 1450-2900 psi (10-20 N/mm ²) Depending on the substrate	ISO 4587 12-24 hrs. cure

After some complications with our bridge building, we had to use a new type of glue to piece the components together. We researched the Tensile Shear strength of the glue to be 10-20 MPA. Since the FOS for this glue is significantly higher, the height of the bridge should decrease to compensate for that increase.

Part I - Introduction

The Matboard Bridge is a box girder modified through Design 0. This design aims to support a small train(weighs 400N with empty load, maximum train weight should be at least 1000N), which is traveling a distance of 1200 mm across the horizontal track from when the first wheels touch to when the last wheels leave. To achieve this, our team (209) embarked on an iterative design process where we addressed challenges such as load distribution at all locations, material limitations, geometry constraints, and ultimately, structural stability. Through extensive experimentation, the design evolved to handle sufficient loading, maintaining the factors of safety necessary against various failure mechanisms. This report outlines the methodology, iterations, and final design decisions that we claim enables the bridge to meet the performance requirements effectively.

In this design, we have decided to prioritize the structural strength over saving material due to several reasons:

1. For civil engineers, the first responsibility is always to ensure the safety of our stakeholders. By doing so, we can cultivate such awareness of safety first, and apply them into practice.
2. It promotes durability and longevity of the design, thereby reducing the future needs for maintenance, resulting in potential economic benefits.

Later in the design process, this principle exerted considerable influence on our design decisions, especially adopting thicker top flanges, etc, eventually leading to our final design. Where the design parameters are as the following:

Parameters & Dimensions (uniform across the bridge):

width top flange	Thickness top flange	Width bottom flange	Thickness bottom flange	Height of web	Thickness of web	Width of the glue	Height of the glue
100	2.54	85	2.54	94.92	1.27	7.5	1.27

It's expected to sustain a maximum weight of train around 1013.4N to travel across a distance of 1200 mm, thereby exceeding the design target.

From Design 0 to the final design, we have many iterations. In those iterations, we made many key design decisions:

1. We installed the diaphragm, which is separated by 400 mm, as a means to tackle shear buckling.
2. We adopted a double-layer top flange in order to better deal with failure due to compression.
3. We decided to place the splice connection at the edge of the bridge, where the shear stress and the bending moment is minimized compared with the central region.
4. We increased the height of the web in order to increase I by increasing area, thereby increasing the FOS compression, FOS shear, and shifting the centroid, potentially resulting in the desired y_{top} / y_{bottom} ratio, as mentioned in design consideration 3.
5. We decided to keep the dimensions uniform across the structure to simplify the construction process.

Part II - Methodology

Material Selection:

In this design, the only material used is the matboard of 1.27mm in thickness and the glue, which is used to bond different components together, safeguarding structural integrity.

Base Design:

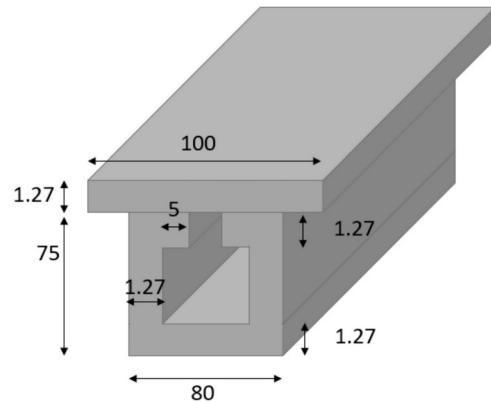
The geometry of Design 0 was chosen as a base case . Matlab was used in conjunction with hand calculation verification to find the maximum shear and maximum bending moment locations to ultimately calculate the factor of safeties against various failure mechanisms. This was achieved by determining forces for load case 1 and load case 2 which is where we iteratively stepped through to our final bridge design, calculating cross sectional properties (areas of individual rectangles, the centroidal axis location, first and second moment of area (I).

The largest shear force occurs when the last train wheels are just beyond the first reaction support pin. Thus, we took into account the fact that 1mm (or as infinitesimally small as we can get) away from that support is where we consider our largest value in the shear force envelope (or when the first wheel is at an arbitrarily small distance away from the further end of the bridge).

Crucial failure mechanisms:

1. Flexural tensile and compression failure.
2. Thin Plate Buckling: calculated the four possible cases: The top plate (Case 1), Overhanging flange on the top plate (Case 2), the web (Case 3), and the side flange (Shear buckling).
3. FOS against: Tension and compression, top plate buckling, side flange buckling, web buckling, shear, glue, and shear buckling.
4. The strength in the splice connection.

Cross-sectional shape and dimensions of Design 0:



Design Considerations:

1. We wanted the centroid to be approximately $\frac{1}{5}$ from top is the best since the ratio between 30 (σ_{tens}) / 6 (σ_{comp}) is 5. It's worth pointing out that this is not necessary but optional.
2. Increasing the thickness of the top plate to shift the centroidal axis upwards.
 - a. Justification: The optimal ratio between the y_{top} and the y_{bottom} is 0.2 due to the aforementioned reason, and the increase in thickness uplifts the area of the top flange,

resulting in the ratio being achieved, as proved by the formula: (used to determine y),
and $y = y_{\text{top}} + y_{\text{bot}}$

3. Increasing the height: this will greatly increase I by increasing area, as proven by $I = \frac{(b \cdot h^3)}{12}$, thereby increasing the FOS compression, FOS shear, and shifting the centroid, potentially resulting in the desired $y_{\text{top}} / y_{\text{bottom}}$ ratio.
4. Since we don't have enough material to avoid such connections, and its location is more fragile in the structure. As a result, we decided to place the connection at the edge of the bridge, where the shear force and the bending moment is minimized compared with the central region.
5. Our group considered adjusting the dimensions of the matboard and maximizing leftover material: Matboard was allocated for diaphragms (made of matboard, thickness is 1.27mm, separated by 400 mm) where the shear forces were the largest. Two on the edge, one in the middle, and the rest are distributed where there were significant jumps in shear force (i.e. the distance between the train wheel point loads).

Iteration Method:

In order to take all the considerations into account, we initially attempted to find an algorithm to directly optimize the solution. However, after conducting research, we discovered that this is infeasible due to the complexity of the structure and the parameters. As a result, we turned to numerical solutions rather than analytical ones. By implementing the core idea of the binary search, we determined the range of the certain parameter (e.g: height of web, thickness of flanges, etc.), then narrowed it down to the optimized value via checking how taking the medium value affects the structure strength, whose results were used to determine how to change the range. We repeat this process until the optimized solution is found.

Part III - Design Proposal

Starting with the design 0, in order to practice the aforementioned principle of safety first, we decided to maximize the structural strength first, then take the material cost into consideration and iterate.

Base Design:

The initial design is of simple structure with uniform dimensions and no extra reinforcements, it is not stable enough to pass the test, revealing the necessity for further iterations.

Major Iterations:

Iteration 1:

Changes Made: Change the height of the web to 197,46 mm from 73.73 mm

Justification: see the aforementioned design consideration 3.

Results: minimum FOS changed from 0.7 to 1.5. The failure case changed from top plate buckling to shear buckling.

Challenges: 1. The strength (failure load = 76N) is still insufficient (maximum train weight is 283.8N). 2. It's going to cost significantly more material.

Iteration 2:

Changes Made: Change the thickness of the web to 2.54 mm from 1.27 mm; change the width of the bottom flange to 65 mm from 80 mm.

Justification: the centroidal axis moves upward as the web gets thicker, resulting in resistance against buckling, as described in design consideration 2.

Results: minimum FOS changed to 2.4. The failure case changed to top plate buckling again.

Challenges: The strength (failure load = 120N) is still insufficient (maximum train weight is 518.3N).

Iteration 3:

Changes Made: Change the thickness of the bottom flange to 0 mm from 2.54 mm; change the width of the bottom flange to 0 mm from 65 mm at location 50 mm and 800 mm from the left support respectively.

Justification: In this iteration, we were trying to see if we can save some material at specific locations. We adopted the idea of varying parameters across the bridge in this iteration.

Results: minimum FOS changed to 1.41. The failure case changed to shear buckling again.

Challenges: The strength (failure load = 70.4N) is still insufficient. This iteration brings only negative effects on the design (maximum train weight of 504.7N)

Iteration 4:

Changes Made: We changed the dimension so that they are uniformly distributed to simplify the building process. We abandoned the last iteration because of its failure to improve on every aspect. Change the height of the web to 114.92.

Justification: In this iteration, we adopted the binary search method with the realization that seeking for an analytical solution is infeasible.

Results: minimum FOS changed to 2.73. The failure case remains as buckling. .

Challenges: The strength (failure load = 136.4N) is still insufficient, but it's approaching our target of 1000N in total (maximum train weight of 977.6N).

Iteration 5:

Changes Made: Change the height of the web to 94.92.

Justification: We conducted another binary search and achieved significant improvement.

Results: minimum FOS changed to 2.83. The failure case changed to compression failure.

Challenges: The strength (failure load = 142.3N) is finally sufficient (the acceptable train weight is now 1013.4N).

Final Design Proposal:

After 5 major and many minor iterations. We now have our final design.

Parameters & Dimensions (uniform across the bridge):

width top flange	Thickness top flange	Width bottom flange	Thickness bottom flange	Height of web	Thickness of web	Width of the glue	Height of the glue
100	2.54	85	2.54	94.92	1.27	7.5	1.27

Expected Performance:

FOS-TEN	FOS-COMP	FOS-CENT	FOS-GLUE	FOS-BUCK 1	FOS-BUCK 2	FOS-BUCK 3	FOS-BUCK V	minFOS	TRAIN FAILURE LOAD	MAX train weight
12.15	2.83	3.82	16.91	5.94	66.9	8.18	3.22	2.83	141.3N	1013 N

Generally speaking, this design meets our requirements by safely supporting a train, which weighs up to 1123N, crossing a distance of 1200 mm. In this design, we drastically changed the dimensions in the base design while still keeping them uniform to simplify the construction process. We implemented diaphragms, which are separated by a distance of 400 mm, as a means to better tackle shear buckling.

FOS-TEN	FOS-COMP	FOS-CENT	FOS-GLUE	FOS-BUCK 1	FOS-BUCK 2	FOS-BUCK 3	FOS-BUCK V	minFOS	TRAIN FAILURE LOAD	MAX train weight
5.55	1.393	1.897	8.35	3.78	12.65	4.41	1.782	1.393	139.3N	999N

Iteration 1	Iteration 2	Iteration 3	Iteration 4	Iteration 5	Iteration 6	Final Design
284 N	518 N	505 N	978 N	1013 N	999 N	999N

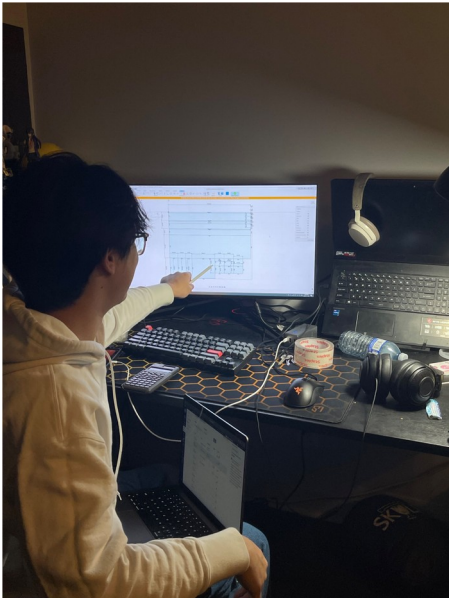
Through these improvements, our final design outperforms the iterations 1 through 4 not only in maximum acceptable train weight (see the form below for the maximum train weight) , but also because

the usage of material needed to be optimized since it was limited. Therefore, it's chosen as our final design.

Part IV - Construction process

In the construction process, many techniques were employed to maximize accuracy of the actual structure. First, we used CAD to carefully plan out to maximize the usage of the material's surface area when cutting and rearranging the matboard.

Pictures:



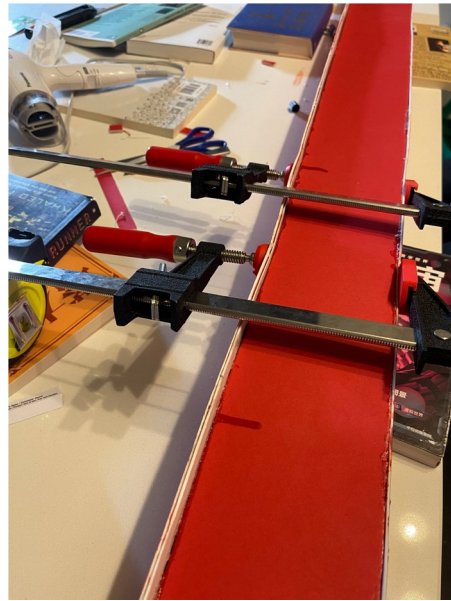
Working together and Brainstorming Ideas



Measuring the Matboard



Bridge Cutting



Bridge Building



Cleaned up workspace

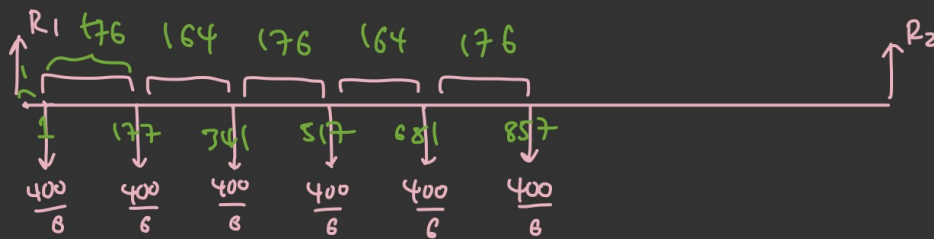
Final Bridge



Design Calculations:

Part V - Hand Calculations

Maximum shear. Train 1mm from edge.



$$\sum M_1 = 0 = -\frac{400}{6} (1 + 177 + 341 + 517 + 681 + 857) + R_2 (1200)$$

$$R_2 = 143.0 \text{ N (up)}$$

$$R_1 = 400 - R_2 = \underline{\underline{257 \text{ N}}} \quad \leftarrow \text{MAX SHEAR FORCE}$$

Maximum moment

R_1

R_2

$\sum M_{i=0} = -\frac{400}{6} ($

$R_1 = 214 \text{ MAX SHEAR}$

147.3 80.7 14

Max bending moment $\sim 69.1 \times 10^3 \text{ Nm}$

Part VI - Code Output Calculations

Load Case 1:

Maximum Bending Moment = 69345.33 occurs at x = 556.00 mm

FOS_tens1 = 4.3684

FOS_comp1 = 1.0390

FOS_cent1 = 2.6704

FOS_glue1 = 9.3985

FOS_buck11 = 0.6177

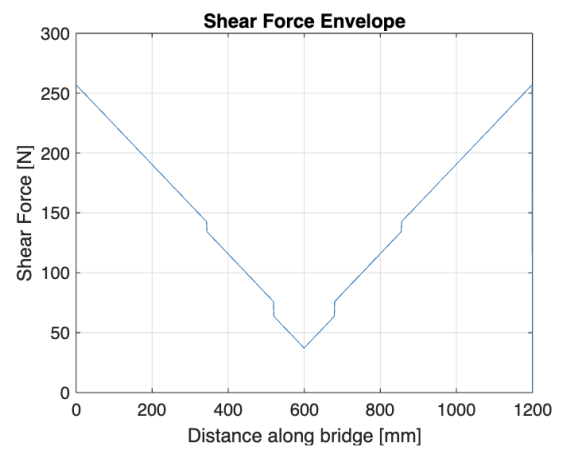
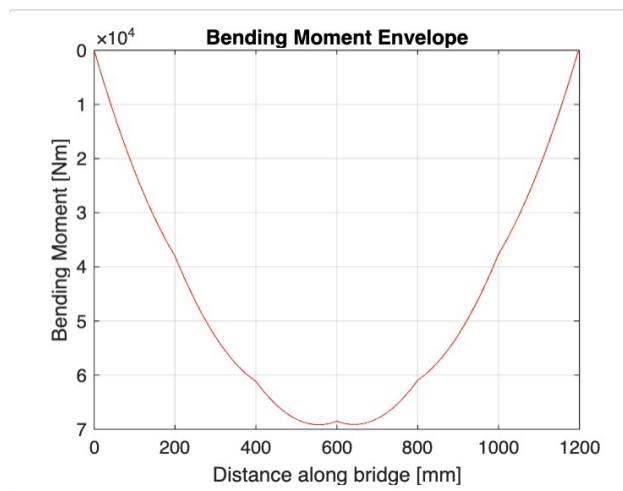
FOS_buck21 = 3.5966

FOS_buck31 = 5.0963

FOS_buckV1 = 3.5094

minFOS = 0.6177

TRAIN_FAILURE_LOAD = 247.0669



Diagrams:

Load Case 2: Final design

Design Calculations

% Cross sectional Properties

```
% Compute the areas of each section
area_top_plate = top_plate_width .* top_plate_thickness
area_bottom_plate = bottom_plate_width .* bottom_plate_thickness
area_web = 2 * (web_height .* web_thickness)
area_glue = 2 * (glue_base_width .* glue_height)

% Compute the centroids of each section
centroid_top_plate = web_height + bottom_plate_thickness + top_plate_thickness / 2
centroid_bottom_plate = bottom_plate_thickness / 2
centroid_web = bottom_plate_thickness + web_height / 2
centroid_glue = bottom_plate_thickness + web_height - glue_height / 2

% Centroidal axis location (centroid_ybar) from the bottom
centroid_ybar = (area_top_plate .* centroid_top_plate + ...
    area_bottom_plate .* centroid_bottom_plate + ...
    area_web .* centroid_web + ...
    area_glue .* centroid_glue) ...
    / (area_top_plate + area_bottom_plate + area_web + area_glue)

% Parallel Axis Theorem: Moments of Inertia
inertia_top_plate = (top_plate_width .* top_plate_thickness.^3 / 12) + ...
    (area_top_plate .* (centroid_top_plate - centroid_ybar).^2)

inertia_bottom_plate = (bottom_plate_width .* bottom_plate_thickness.^3 / 12) + ...
    (area_bottom_plate .* (centroid_bottom_plate - centroid_ybar).^2)

inertia_web = 2 * ((web_thickness .* web_height.^3 / 12) + ...
    (area_web / 2 .* (centroid_web - centroid_ybar).^2))
```

```
area_top_plate = 1x1201
    254    254    254    254    254 ...

area_bottom_plate = 1x1201
    190.5000    190.5000    190.5000 ...

area_web = 1x1201
    241.0968    241.0968    241.0968 ...

area_glue = 1x1201
    19.0500    19.0500    19.0500 ...

centroid_top_plate = 1x1201
    98.7300    98.7300    98.7300 ...

centroid_bottom_plate = 1x1201
    1.2700    1.2700    1.2700 ...

centroid_web = 1x1201
    50    50    50    50    50 ...

centroid_glue = 1x1201
    96.8250    96.8250    96.8250 ...

centroid_ybar = 55.6573
inertia_top_plate = 1x1201
    10^5 x    4.7137    4.7137    4.7137 ...
```

```
% First Moment of Area at Glue Location
Q_glue_ft = first_moment_top_plate
```

```
% Distance from centroid to extremes
distance_to_top = top_plate_thickness + web_height + bottom_plate_thickness - centroid_ybar
distance_to_bottom = centroid_ybar
```

```
% Ratio of distances for flexural strength
ratio_flexural = distance_to_top / distance_to_bottom
```

```
Q_glue_ft = 1x1201
10^4 × 1.0940 1.0940 1.0940 ...
```

```
distance_to_top = 1x1201
44.3427 44.3427 44.3427 ...
```

```
distance_to_bottom = 55.6573
ratio_flexural = 1x1201
0.7967 0.7967 0.7967 ...
```

```
0.7967 0.7967 0.7967 ...
```

% Stresses

```
% Calculate bending stresses at the top and bottom of the bridge
stress_top_plate = abs(BMDmax) .* (top_plate_thickness + web_height + bottom_plate_thick
% Stress at the bottom plate due to bending
stress_bottom_plate = abs(BMDmax) .* centroid_ybar ./ total_inertia
```

```
% Calculate shear stress at the center of the web
shear_stress_web_center = abs(SFDmax) .* total_first_moment_area ./ (total_inertia .* we
```

```
% Calculate shear stress at the glue locations
shear_stress_glue_location = abs(SFDmax) .* Q_glue_ft ./ ...
(total_inertia .* (2 * glue_base_width + 2 * 1.27))
```

```
stress_top_plate = 1x1201
0 0.0155 0.0309 ...
```

```
stress_bottom_plate = 1x1201
0 0.0194 0.0388 ...
```

```
shear_stress_web_center = 1x1201
1.9182 1.9156 1.9130 ...
```

```
shear_stress_glue_location = 1x1201
0.2179 0.2176 0.2173 ...
```

% Thin plate buckling

```
% Case 1: Buckling of the top flange
effective_top_flange_width = top_plate_width - (top_plate_width - bottom_plate_width) - web_thick
critical_stress_case1 = (4 * pi^2 * E) / (12 * (1 - mu^2)) * (top_plate_thickness / effective_top
```

```
critical_stress_case1 = 16.2685
```

```
% Case 2: Buckling of the remaining portion of the top flange
remaining_flange_width = (top_plate_width - effective_top_flange_width) / 2;
critical_stress_case2 = (0.425 * pi^2 * E) / (12 * (1 - mu^2)) * (top_plate_thickness / remaining
```

```
critical_stress_case2 = 54.4632
```

```
% Case 3: Web buckling
web_effective_thickness = web_height + bottom_plate_thickness - centroid_ybar;
web_buckling_stress = (6 * pi^2 * E) / (12 * (1 - mu^2)) * (web_effective_thickness / web_thickne
```

```
web_buckling_stress = 18.9783
```

```
% Case 4: Shear buckling
diaphragm_spacing = 200; % Distance between diaphragms
shear_buckling_stress = (5 * pi^2 * E) / (12 * (1 - mu^2)) * ((web_thickness / web_height)^2 + (w
```

```
shear_buckling_stress = 1x1201
3.7583 3.7583 ...
```

FOS Calculations

```
% Tension and compression
FOS_tens = tensile_stress ./ stress_bottom_plate; % FOS for tensile stress
FOS_comp = compressive_stress ./ stress_top_plate; % FOS for compressive stress
```

```
allFOS = 1x8
5.5499 1.3932 ...
```

```
% Shear
FOS_shear = max_shear_stress ./ shear_stress_web_center; % FOS for shear stress at the web cent
FOS_glue = max_glue_shear_stress ./ shear_stress_glue_location; % FOS for glue shear stress
```

```
min_FOS = 1.3932
```

```
FOS_tens1 = 5.5499
```

```
FOS_comp1 = 1.3932
```

```
FOS_cent1 = 1.8967
```

```
FOS_glue1 = 8.3469
```

```
FOS_buck11 = 3.7775
```

```
FOS_buck21 = 12.6464
```

```
FOS_buck31 = 4.4068
```

```
FOS_buckV1 = 1.7821
```

```
% Buckling
FOS_buck1 = critical_stress_case1 ./ stress_top_plate; % FOS for top plate buckling case 1
FOS_buck2 = critical_stress_case2 ./ stress_top_plate; % FOS for top plate buckling case 2
FOS_buck3 = web_buckling_stress ./ stress_top_plate; % FOS for web buckling
FOS_buckV = shear_buckling_stress ./ shear_stress_web_center; % FOS for shear buckling
```

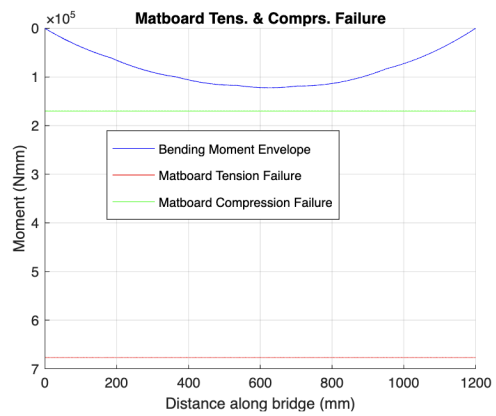
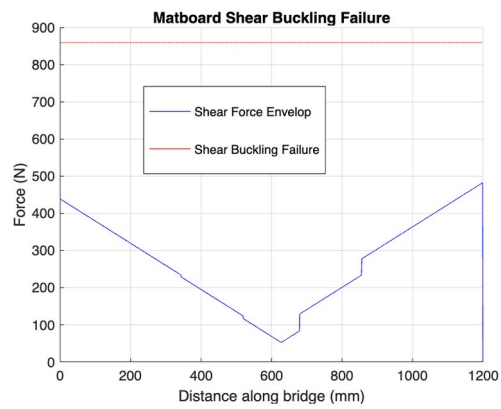
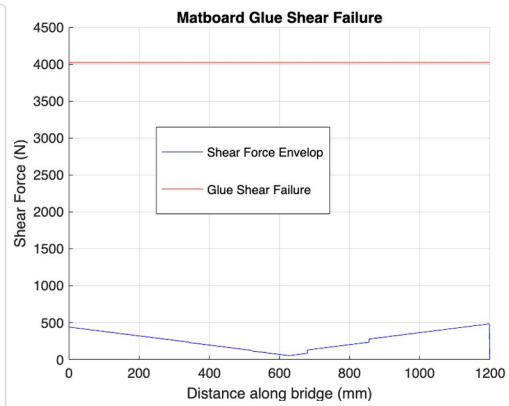
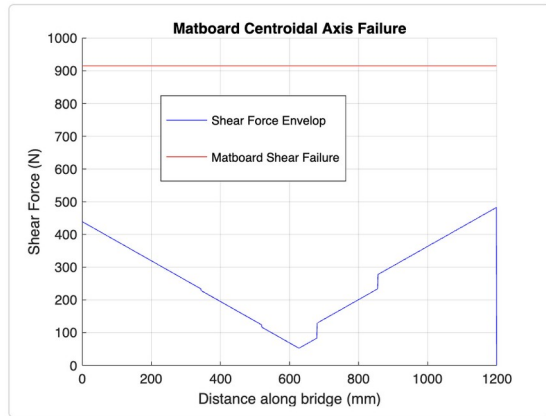
```
% Identify the minimum FOS for critical failure prediction
minFOS = [min(FOS_tens), min(FOS_comp), min(FOS_shear), ...
min(FOS_glue), min(FOS_buck1), min(FOS_buck2), ...
min(FOS_buck3), min(FOS_buckV)];
```

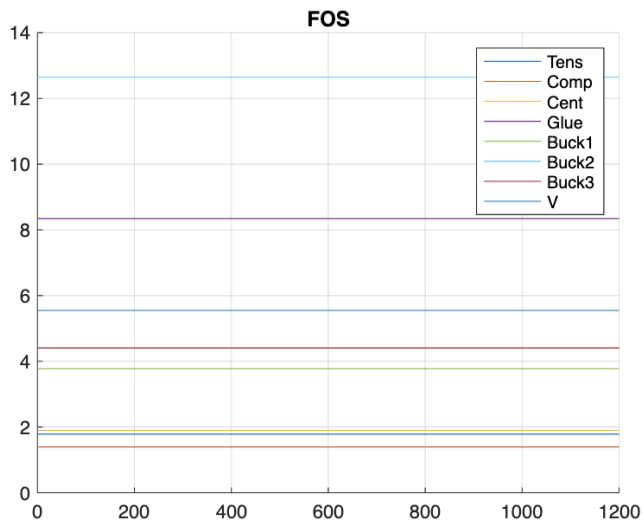
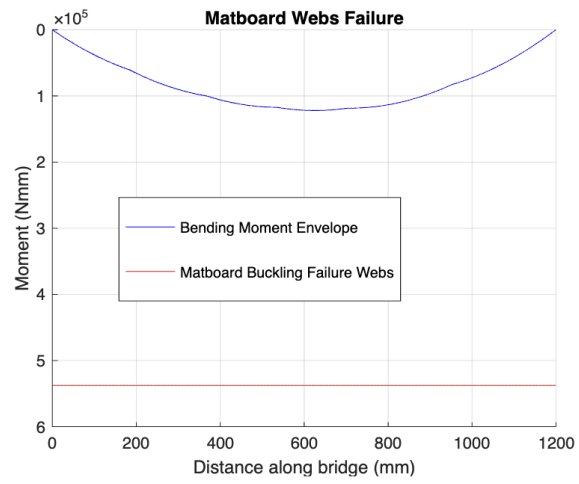
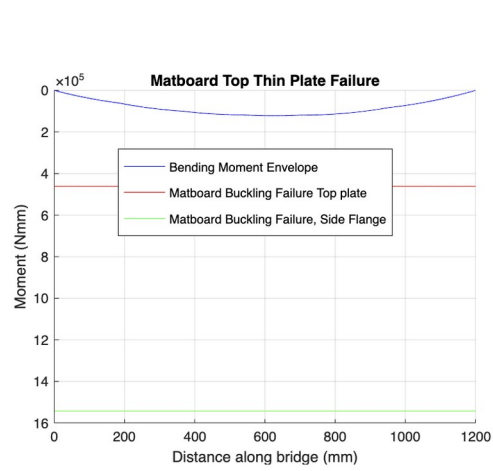
```
allFOS = [min(FOS_tens), min(FOS_comp), min(FOS_shear), min(FOS_glue), min(FOS_buck1), min(FOS
min_FOS = min(allFOS)
FOS_smaller_than_1 = allFOS(allFOS<1);
```

```
TRAIN_FAILURE_LOAD = 139.3200
```

```
PPP_FAIL_TOTAL_LOAD = 998.9243
```

```
FOS_tens1 = allFOS(1)
FOS_comp1 = allFOS(2)
FOS_cent1 = allFOS(3)
FOS_glue1 = allFOS(4)
FOS_buck11 = allFOS(5)
FOS_buck21 = allFOS(6)
```





Part VII - Potential Improvements

Despite the success of the final design, there are still several potential improvements to be made.

1. When making the diaphragms, we had to dig triangular slots so that their edges can be perpendicular to the diaphragms, thereby allowing the use of glue to fix the diaphragms. However, this process is lengthy and hard to maintain with precision, leading to deviation of performance from the expected level, thereby revealing the necessity of exploring an improved methodology of producing the diaphragms.
2. It's also valuable, in the long term, to further explore the possibilities of continuing improving the quality of the codes, and possibly, find the algorithms to optimize the solutions, thereby saving much of our time, resulting in better efficiency.
3. Cutting out the triangles along the edges contributed to a large uncertainty in our measurements. We wish to find a method to improve the accuracy of our cutting.

Part VIII - Conclusion

In this project of designing a bridge, in order to sustain a train with various load cases to travel a distance of 1200 mm, we have developed a final design meeting these criterias, with the iterative strategy and rigorously analysis. We prioritized the structural strength over material saving, resulting in sustaining the maximum train weight of 999 N, with significant improvements being made on the basis of Design 0, such as implementing diaphragms, adopting double-layer top flanges, and drastically changing the dimensions. From this experience, we also learned the possibilities of improving the production and analysis methods to boost precision and efficiency.

References

[1]“Super Glue Longneck Bottle,” *Lepage.ca*, 2019.
<https://www.lepage.ca/en/lepage-products/fix-stuff/super-glue/liquid-super-glue/super-glue-long-neck.html> (accessed Nov. 26, 2024).

```
clear; close all;
```

% Initial Parameter setting

```
L = 1200; % Length of the bridge, in mm  
DeltaL = 1; % 1 mm per segment
```

Team 209 Time Log

Tasks	Date	Ethan	Amy	Xin	Total
REPORT					
Added design step	Nov 21	0.5	1	4	
Introduction	Nov 21	0.5	1	1	
Design iterations	Nov 21	1		3	
Potential improvements	Nov 23-24			0.5	
Conclusion	Nov 23-24			0.5	
					13
CALCULATION	Nov 21	0.5	2.5	0.5	
					3.5
DRAWINGS					
Elevation, sectional views	Nov 24-26	4			
Construction plans	Nov 23-24	2			
Iteration drawings with different dimensions	Nov 23			1	
Creating visual MAT board layout component allocation sketch	Nov 23	1			
CAD - ding the Bridge in Fusion	Nov 23	3			
					13
CONSTRUCTION	Nov 23-24 All				
Moving the allocation sketch from AutoCAD on matboard		3	3		
Design construction procedure due to diaphragms		1			

Detailed cutting plan marked on cutted pieces		2	2		
Cutting the main pieces of the bridge		3			
Cutting the diaphragms		1			
Glueing bottom flange and top flange		0.1	0.1	0.1	
Glueing side to bottom flange		0.1			
Clamping		2	3	3	
Glueing diaphragm		0.25	0.25	0.25	
Accident: bridge collapsed - re-glueing diaphragms		0.25	0.25	0.25	
					24.9
OTHERS					
code parameters	Nov 20-25		7		
optimization code	Nov 23-24		3	5	
Excel keep track and optimization	Nov 23-24			2	
Iterations and coding and optimization	Nov 23-24	6		3	
Record process	Nov 23-24	2	2	2	
					32
TOTAL		33.2	25.1	26.1	86.4

“Signatures”:

Amy L.

Ethan L.

Xin Wang

Architectural floor plan showing dimensions in meters. The plan includes a large rectangular area with a central corridor and several rooms. Dimensions are provided for the overall footprint, room sizes, and corridor widths. The total width is 1000.00m and the total length is 285.00m. The plan includes a north arrow pointing towards the top right.

Overall dimensions:

- Total width: 1000.00m
- Total length: 285.00m

Room dimensions (from left to right):

- Room 1: 250.00m x 100.00m
- Room 2: 250.00m x 100.00m
- Room 3: 250.00m x 75.00m
- Room 4: 250.00m x 115.00m
- Room 5: 250.00m x 115.00m
- Room 6: 250.00m x 115.00m

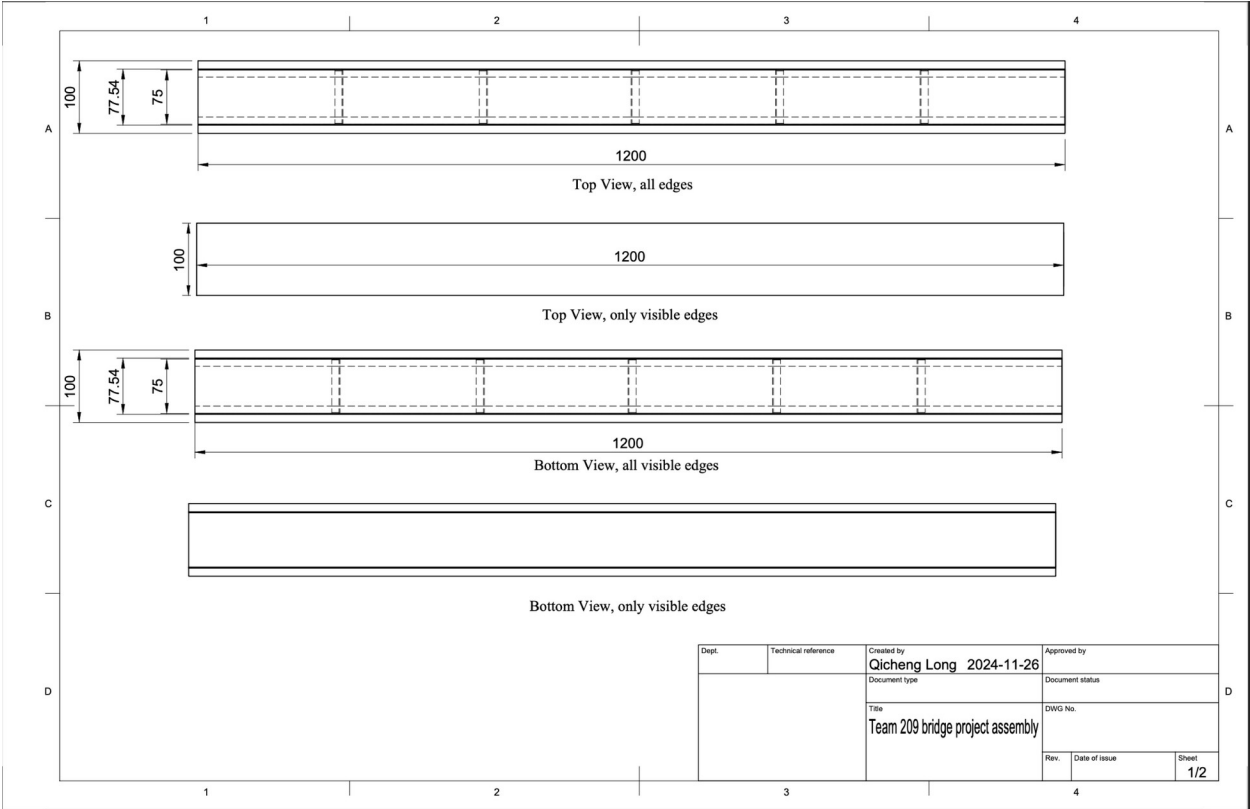
Corridor dimensions (from left to right):

- Corridor 1: 75.00m wide
- Corridor 2: 100.00m wide
- Corridor 3: 100.00m wide
- Corridor 4: 100.00m wide
- Corridor 5: 100.00m wide
- Corridor 6: 100.00m wide
- Corridor 7: 100.00m wide
- Corridor 8: 100.00m wide
- Corridor 9: 100.00m wide
- Corridor 10: 100.00m wide
- Corridor 11: 100.00m wide
- Corridor 12: 100.00m wide
- Corridor 13: 100.00m wide
- Corridor 14: 100.00m wide
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- Corridor 99: 100.00m wide
- Corridor 100: 100.00m wide

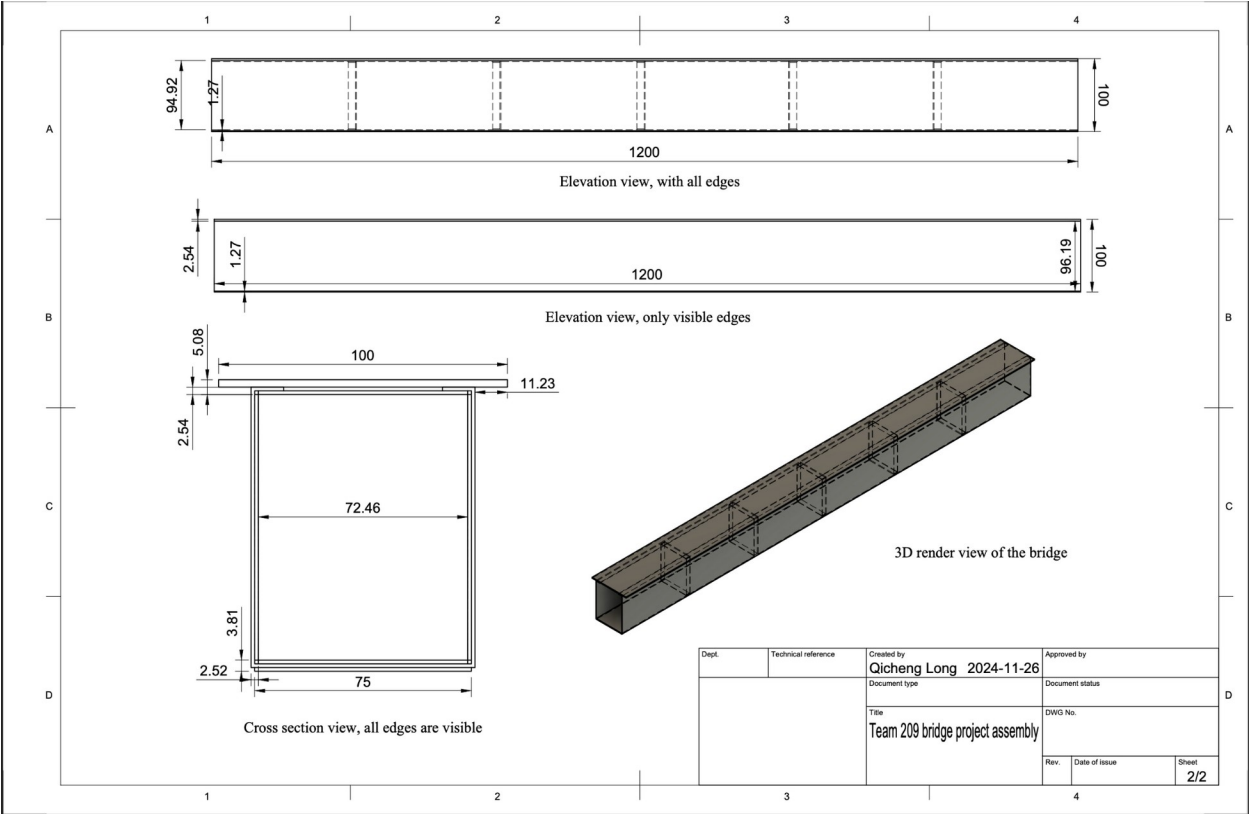
Fig.1 MAT board cutting plan

Team 209 Engineering Drawings

All important drawings are to scale of 1:4 except cross section view for a better visualization and clearer dimensions



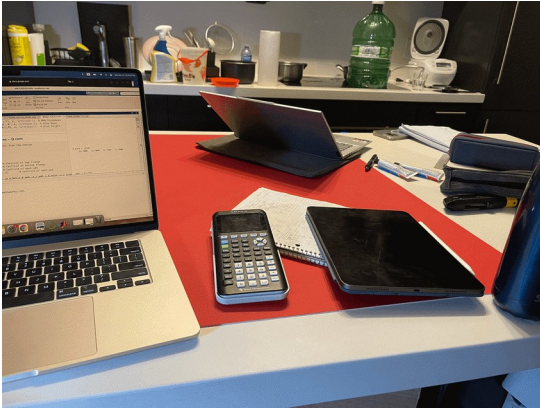
All important drawings are to scale of 1:4 except cross section view for a better visualization and clearer dimensions



[Please see next page]

Team 209 Bridge Building Process

Design and Merging:



[Nov 23 | 6:30 PM]

Description: Writing final code and preparing for optimization calculations



[Nov 23 | 11:02 PM]

Description: Pointing at matboard piece allocation layout (CAD)



[Nov 23 | 11:01 PM]

Description: Marking some points on matboard to know where to cut

Cutting:



[Nov 24 | 12:33 AM]

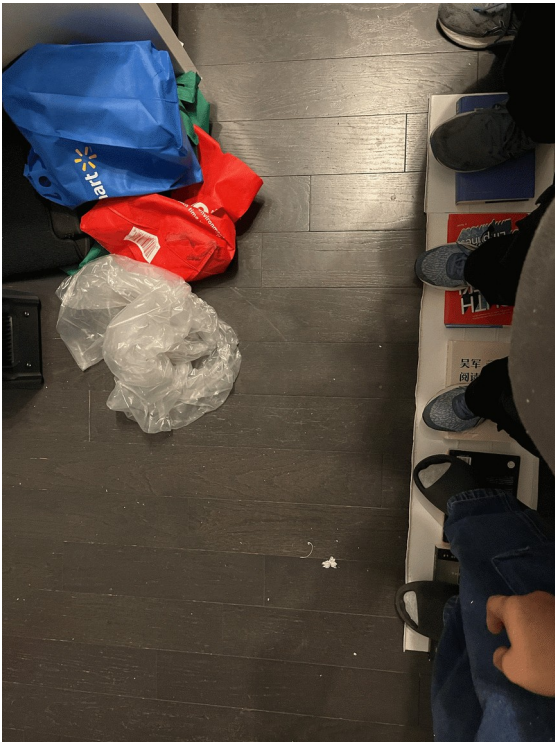
Description: Beginning to cut the matboard along the liens we drew



[Nov 24 | 3:58 AM]

Finished cut pieces

Glue drying:



[Nov 24 | 5:58 AM]

Standing on Matboard to dry the glue

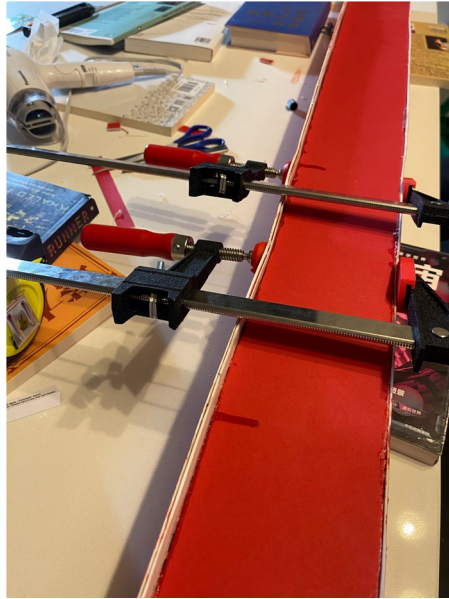


[Nov 24 | 6:25 AM]

Holding down the diaphragms



[Nov 24 | 5:07 PM]
Diaphragms and bridge inside



[Nov 24 | 6:08 PM]
Clamps to hold final bridge together



[Nov 24 | 7:00 PM]
Final bridge

To keep our workspace clean during the process of the bridge construction, we collected the waste materials into a garbage bag and gathered the scraped paper (scraped out from the corner of the MATboard to form a nice 90 degree angle) into a corner on the floor for us to clean up later

Cleaning up:



[Nov 24 | 7:10 PM]
Cleaned up workspace

To clean up, Ethan and Xin mopped the floor and Ami cleaned up the workspace.